Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION
AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Worksheet No. 3 | Week 6: Computer Networks & the Internet (CO4)

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

Name: Yreignoa H. Navarro **Block:** BSCS 1-1

Computer Unit Used (Desktop/Laptop, brand): **LENOVO** **Date:** 10/16/26

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for

example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled or disconnected.

3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.

8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost

9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com

10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4

11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:

Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

Layer 3 – NetworkV. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: _____

Adapter Name	Description	Physical Address (MAC)	Status
Wi-Fi	Intel(R) Wi-Fi 6 AX201 160MHz	3C-9C-0F-12-34-56	Connected
Ethernet	Realtek PCIe GbE Family Controller	00-1A-2B-3C-4D-5E	Disconnected
Bluetooth Network Connection	Bluetooth Device (Personal Area Network)	3C-9C-0F-12-34-57	Disconnected
Local Area Connection* 1	Microsoft Wi-Fi Direct Virtual Adapter	3E-9C-0F-12-34-58	Disconnected

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	**Item Checked (continued)**
Client for Microsoft Networks	QoS Packet Scheduler
File and Printer Sharing for Microsoft Networks	Internet Protocol Version 4 (TCP/IPv4)
Internet Protocol Version 6 (TCP/IPv6)	Microsoft LLDP Protocol Driver
Link-Layer Topology Discovery Mapper I/O Driver	Link-Layer Topology Discovery Responder

TCP/IPv4 Setting	**Automatic or Static?**
IP address	Automatic (DHCP)
DNS server address	Automatic

Table 3. Command Prompt Results (Part C)

Command / Item	**Your Findings**	**OSI Layer**
hostname	DESKTOP-4MKSC1R	Layer 7 – Application
getmac /v: MAC address and Transport Name	MAC: 3C-9C-0F-12-34-56 Transport name: 9B4D03C-8E12-4C50-96D0-B7C2C5D36CD2	Layer 2 – Data Link
ipconfig /all: Media State of Ethernet	Media disconnected	—
ipconfig /all: DHCP Enabled	Yes	—
ping 127.0.0.1: replies and time	Reply from 127.0.0.1: bytes=32 time<1ms TTL=128	Layer 3 – Network
ping localhost: address that replied	127.0.0.1	—
ping 8.8.8.8: exact message	Reply from 8.8.8.8: bytes=32 time=20ms TTL=117	Layer 3 – Network
nslookup google.com: exact message	Name: google.com followed by one or more IP addresses	Layer 7 – Application
netstat -an: three LISTENING addresses:ports	0.0.0.0:135; 0.0.0.0:445: 127.0.0.1:49664	Layer 4 – Transport
route print -4: Netmask for 127.0.0.0	255.0.0.0	—
systeminfo: number of NICs and status	3 NICs Installed — Wi-Fi connected; Ethernet and Bluetooth disconnected	—

Table 4. PowerShell Results (Part D)

Command	**What It Showed**	**Matches CMD? (Y/N)**
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	Wi-Fi — Up; MAC 3C-9C-0F-12-34-56; Link speed 866.7 Mbps. Ethernet — Disconnected.	Y
Get-NetIPAddress: IPv4 addresses and PrefixLength	Wi-Fi — 192.168.1.10/24; PrefixLength 24	Y
Test-Connection 127.0.0.1: result	Successful replies from 127.0.0.1	Y
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	0.0.0.0:135; 0.0.0.0:445	Y

Table 5. Sample Output Analysis (Part E)

Question	**Your Answer**
IPv4 address of PC07	
Subnet mask (and its /prefix)	255.255.255.0 (/24)
Default gateway, and what device it usually is	192.168.10.1
DNS servers	8.8.8.8 and 8.8.4.4
Was the address assigned by DHCP or set manually? How do you know?	DHCP, if the computer's settings show DHCP enabled and automatic IP assignment.
Public or private? Which RFC 1918 range?	Private IPv4 address; 192.168.10.7 belongs to the 192.168.0.0/16 range.
IPv4 address of PC07 in binary (four octets)	11000000.10101000.00001010.00000111
MAC address, and the OSI layer where it is used	3C-9C-0F-12-34-56
How long is the DHCP lease?	Depends on the DHCP server; commonly 24 hours
If the DHCP server failed, what kind of address would PC07 assign itself?	An APIPA address in 169.254.0.0/16, usually with subnet mask 255.255.0.0.

Figure 1. Proposed Laboratory Network Diagram (Part F)

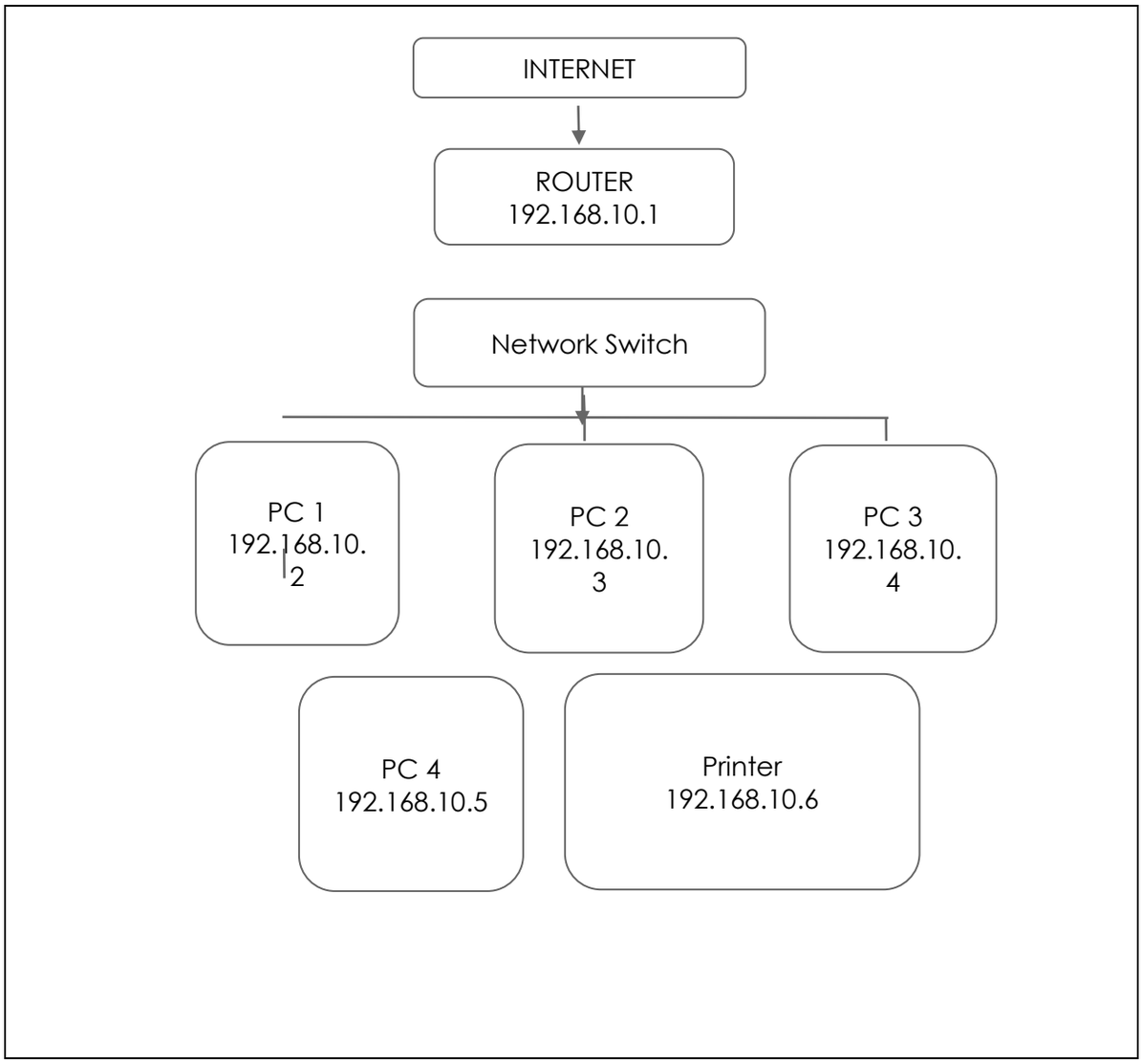


Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.2	255.255.255.0	192.168.10.1
PC 2	192.168.10.2	255.255.255.0	192.168.10.1
PC 3	192.168.10.2	255.255.255.0	192.168.10.1
PC 4	192.168.10.2	255.255.255.0	192.168.10.1
Printer	192.168.10.2	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

The hosts file maps hostnames to IP addresses on a computer. It can override DNS lookups for specific hostnames without requiring a DNS server.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have? Why do you think it still has it?

Even without a network connection, a computer still has a hostname, MAC address, and network adapter information. These details are stored locally on the computer. They help identify the device and its network interfaces.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

I found Command Prompt most useful because it provides detailed network information through simple commands. It can display IP addresses, MAC addresses, and connectivity results. It is also useful for troubleshooting network problems.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

The ping 8.8.8.8 command checks whether the computer can reach Google's public DNS server by IP address. The nslookup google.com command checks whether the domain name can be resolved to an IP address. These commands help identify possible connectivity or DNS problems.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

We should be careful because these details can reveal information about our devices and network configuration. Sharing them publicly may expose information that could help others target a network. Protecting this information helps maintain privacy and network security.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30

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Criteria	Description	Points
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**